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GEO-Insight

Which humanitarian crises are most overlooked?

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Repository: github.com/BenBullinger/GEO-Insight

THE QUESTION

Relative to documented need, which crises are *most overlooked*?

Humanitarian coordinators and donor advisors need a defensible answer every cycle. No existing tool gives them one.

The scale of the gap

$33B

2026 GHO appeal
to reach 135M people

$12B

Raised in 2025
the lowest in a decade

19

Severely underfunded contexts
up from 8 in 2021

Sudan fell from 69 % covered (2024) to 35 % (2025). Yemen, Somalia, and DRC sit near 22–24 %.

THE ANALYTICAL PROBLEM

When *every* response is underfunded, "underfunded" stops being useful.

A single coverage ratio can't discriminate between 19 contexts that are all below 50 %. The interesting variation is no longer *whether* a crisis is underfunded — but *how*, and *for whom*.

PART I

A sharper question.

Three ways to be overlooked

Existing tools collapse these into one number. They aren't the same crisis.

HIGH NEED · LOW FUNDING · LOW ATTENTION

Forgotten

Angola drought, 2023 — 7M affected. Just 1,049 news articles worldwide. (*Barbie*, same year: 273,279.)

HIGH NEED · LOW FUNDING · HIGH ATTENTION

Contested

Sudan, 2025 — world's largest displacement crisis, on every front page. Coverage still dropped from 69 % to 35 %.

ADEQUATE AGGREGATE · STARVED CLUSTER

Sector-starved

A crisis at 60 % aggregate coverage can hide a **health cluster at 20 %** while food is overfunded. The total masks the starved part.

From one number to a joint distribution

EXISTING TOOLS

0.37

A scalar coverage ratio.
Cannot distinguish the three kinds.

→

GEO-INSIGHT

intra-crisis equity
× inter-stakeholder disagreement

over a six-signal score

A distribution that distinguishes *forgotten*, *contested*, and *sector-starved*.

Each new axis is anchored in an explicit gap in the 2015–2026 humanitarian-analytics literature (Abdulrashid 2026 §7.3; Sahebjamnia 2017 §6; Vargas Florez 2015 Eqs. 9–12).

How the score works

For each crisis, **six signals** are weighted by a **donor profile** and aggregated over *observed* attributes:

$$G_p(u) \;=\; \sum_{i \in \mathcal{O}(u)} w_{p,i} \cdot U_i(\tilde{a}_i(u))$$

$G_p(u)$ gap score for crisis u under profile p

$\mathcal{O}(u)$ indices of attributes observed for u

$w_{p,i}$ profile p's weight on signal i

$U_i(\cdot)$ utility transform for signal i

$\tilde{a}_i(u)$ min-max normalised signal value

Six signals: coverage shortfall · per-PIN funding gap · need intensity · severity · donor concentration · *intra-crisis equity Gini*. Four profiles: CERF-, ECHO-, USAID-, NGO-flavoured.

Evidence-bounded: missing attributes contribute zero (no renormalisation). A partial-complete country's max score is bounded by its observed weight-mass — incompleteness cannot inflate rank.

Sources: 5 OCHA datasets (HNO, HRP, FTS, CoD-PS, CBPF); severity from INFORM (EU JRC, monthly, 109 countries).

Four kinds of overlooked, decoded

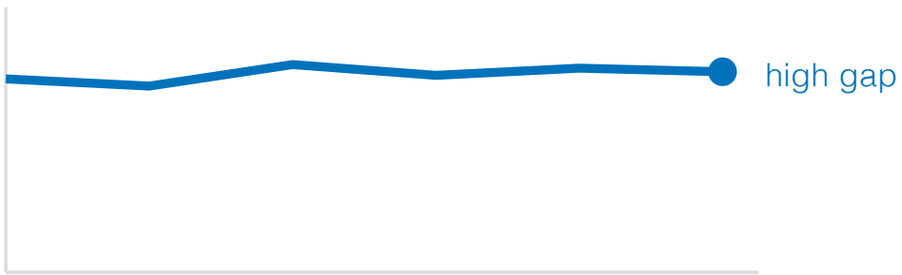
Rank each crisis under *every* donor profile. Combine the spread of ranks with the sector inequality.

	LOW DISAGREEMENT	HIGH DISAGREEMENT
HIGH EQUITY GAP	**Consensus-starved-sector** Clearest operational target. One cluster missing; every mandate agrees.	**Contested *and* sector-starved** Highest advocacy priority. Politically contested AND a missing cluster.
LOW EQUITY GAP	**Consensus-overlooked** CERF UFE candidates. Everyone agrees — underfunded across the board.	**Contested but balanced** Reveals donor politics. Priorities diverge; no missing cluster.

A third axis: time

A crisis overlooked for five years needs different policy than one that just tipped. We separate the two.

Chronic



Persistent underfunding across years. **Response:** multi-year pooled funding.

Acute

Recent collapse from baseline. **Response:** CERF rapid-response window.

Chronic = k-year rolling mean of the gap. Acute = deviation of this year from that rolling mean.

Robustness to incomplete data

Most crises lack at least one of the six attributes. Three candidate handlings, one is defensible.

HARD-DROP

Wasteful

Only **18 / 114** eligible countries scored. The other 96 drop out silently.

REWEIGHT OBSERVED

Inflates ranks

Dividing out missing-weight mass lets partial-complete countries with extreme observed attributes **displace** the true top. On our CERF UFE validation: top-5 overlap collapses **100% → 0%**.

EVIDENCE-BOUNDED · OUR CHOICE

Robust

Missing ≡ 0 contribution. Max score bounded by observed weight-mass. Pool 57 preserves **80%** CERF overlap — within one country of the strict pool.

This choice follows directly from equation (1) on the previous slide: we sum, we do not renormalise. The cost — systematic under-ranking of poorly-documented crises — is the accepted flip side of refusing to extrapolate beyond observation.

PART II

Does it agree with reality?

We don't validate against ourselves

Two lists, human-curated by independent organisations, not using our method:

OCHA

CERF Underfunded Emergencies

Twice-yearly allocation to "*protracted or neglected humanitarian crises where funding is low but vulnerability is high.*" 17 countries selected in 2025 across both windows; 22 across 2024–25 combined.

→ validates our *underfunding* axis.

CARE

Breaking the Silence 2024

Annual ranked top-10 of the most *under-reported* humanitarian crises. Measured by media attention — not by funding.

→ validates our *sector-starved / forgotten* axis (partially, by design — different signal).

Metrics: precision at top-K · recall of benchmark · Spearman ρ on the intersection · stability across pool sizes.

Top-10 — evidence-bounded, All pool (n = 57)

#	ISO3	Country	Median rank	IQR	Completeness	Typology
1	**SDN**	Sudan	2.5	3.50	1.0	consensus-sector-starved
2	**HND**	Honduras	2.5	3.25	1.0	consensus-sector-starved
3	**AFG**	Afghanistan	5.0	1.00	1.0	consensus-overlooked
4	ZWE	Zimbabwe	5.0	8.75	0.67	contested-sector-starved
5	**SOM**	Somalia	5.5	4.00	1.0	consensus-overlooked
6	**HTI**	Haiti	6.0	2.50	1.0	consensus-sector-starved
7	**MWI**	Malawi	7.5	12.00	0.67	contested-sector-starved
8	**MOZ**	Mozambique	9.0	4.50	1.0	consensus-overlooked
9	SSD	South Sudan	9.0	1.75	1.0	consensus-overlooked
10	**TCD**	Chad	9.5	7.75	1.0	contested-balanced

Bold = also in OCHA's CERF Underfunded Emergencies 2024–25. 8 of 10 overlap. Orange completeness = partial observation (4 of 6 attributes); ranked alongside strict-complete countries under evidence-bounded scoring.
Low IQR (AFG 1.00, SSD 1.75) ⇒ consensus across donor profiles. High IQR (MWI 12.00) ⇒ contested.

Validation against external benchmarks

Pool (n countries)	CERF UFE 2024–25 (22)	CERF UFE 2025 only (17)	CARE BTS 2024 (10)
Strict (c=1.0, n=18)	Overlap@5 = **100 %** · @10 = 80 %	@5 = **100 %** · @10 = 80 %	@10 = 10 % · ρ = +0.40
High (c≥2/3, n=33)	@5 = 80 % · @10 = 80 %	@5 = 80 % · @10 = 70 %	@10 = 20 % · ρ = –0.11
All (c≥1/2, n=57)	@5 = **80 %** · @10 = **80 %**	@5 = 80 % · @10 = 70 %	@10 = 20 % · ρ = –0.43

Headline. Tripling the pool (18 → 57) by admitting partial-complete countries costs *one* country at top-5 against CERF UFE (100 % → 80 %). The methodology is robust to incompleteness.

CARE Spearman ρ is methodologically expected to drift as pool grows: CARE ranks *media attention*, not funding. High-attention crises (Sudan) top our ranking but are by definition not "under-reported" — they drop out of CARE.

What this is *not*

NOT AUTOMATION

The tool ranks. Humans decide. A score is a hypothesis, never a verdict.

NOT A SINGLE NUMBER

Four donor profiles, four ranks per crisis. We report the median *and* the IQR.

NOT IMPUTED

Missing attributes contribute zero to the score, never country-specific guesses.

NOT NEUTRAL

We *surface* donor disagreement as signal instead of hiding it in consensus.

NOT COMPLETE

Completeness flag attached to every score. Evidence-bounded scoring cannot inflate partial-complete ranks.

NOT PRECISE

Two significant figures. A score of 0.62 is not meaningfully distinct from 0.59.

THANK YOU

Questions?

`proposal/proposal.pdf` · `github.com/BenBullinger/GEO-Insight`